

Macroeconomic Effects of Foreign Direct Investment: Should We Worry about Offshoring?

Shinnosuke Kikuchi Ippei Fujiwara

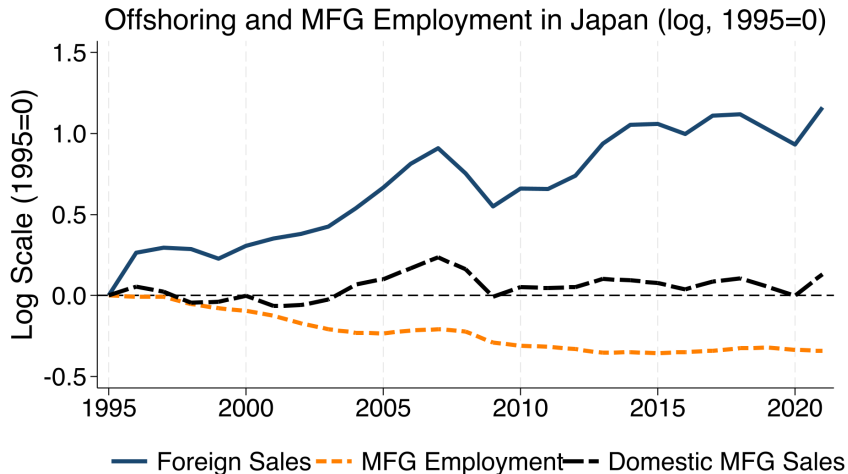
July 5, 2026

Kyoto Summer Workshop on Applied Economics 2026

Doshisha University

Preliminary; working paper expected in 5 weeks

Rise of Offshoring, Decline of Manufacturing Employment



Source: BSOBA, BSJBSA, JIP



More Offshoring, Fewer Jobs? Should We Worry?

- Important and Controversial Topic
 - “One of the **most controversial** aspects of globalisation is offshoring.”
 - Kiyota et al (2022, VOXEU)
- “**little agreement** among academic economists regarding the sign...”
 - Kovak et al (2021, RESTAT)
- Some evidence (in the US or Japan)
 - Parents' overall emp: ↑: Kovak et al (2021, RESTAT),...
 - Inequality: ↑: Feenstra and Hanson (1999, QJE), Hummels et al (2014, AER)
 - Regional/Sectoral emp: zero or ↑? Kovak et al (2021), Kiyota et al (2025)

This Paper

RQ: What is the micro/macro effect of FDI on labor, productivity, and welfare?

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- Employment
 1. Micro: Across Firms – Employment increases.
 2. **Macro: MFG Aggregate (Japan) – Employment decreases**
 - Confirm sizable substitution against domestic suppliers
 - Simple PE accounting: Net job loss of about 399K (rel. 1.26M affiliate jobs)
 - GE: exposed-sector loss remains negative (-688K), offset by nontradables
- GE: labor reallocates to nontradables; aggregate TFP falls; welfare rises (???)

Firm-level Evidence

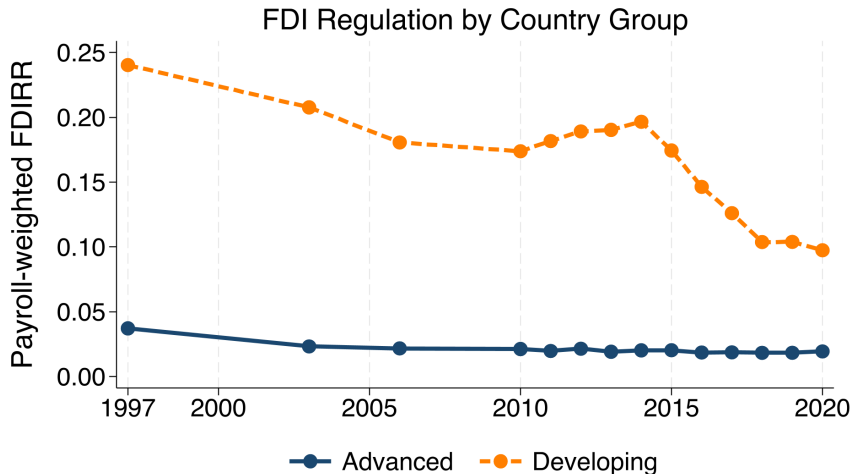
Offshoring and Employment

- Firm-level changes in employment on changes in offshoring

$$\Delta \ln L_i = \beta \Delta \text{OFS}_i + \varepsilon_i$$

- ΔOFS_i : Changes in firm i 's foreign affiliates' employment (BSOBA)
- Of course, this is problematic—Omitted Variable Bias
 - Positive demand/supply shock to the firm can increase both L and OFS

FDI Regulation Index by Country



Source: BSOBA, OECD

Foreign Affiliates by Host Country, 2019

Host country	Affiliates	%	Employment	%
China	5,240	26.4	1,254,477	22.3
United States	2,193	11.0	749,447	13.3
Thailand	1,960	9.9	641,793	11.4
Indonesia	965	4.9	402,053	7.1
Vietnam	956	4.8	389,887	6.9
Total	19,885	100.0	5,626,167	100.0

Source: BSOBA

Firm-level Analysis: Specification

- Long-difference between 2010-2019; with Shift-Share IV (SSIV); i firms

$$\Delta \ln L_i = \beta \Delta \text{OFS}_i + \varepsilon_i$$

- ΔOFS_i : Changes in foreign affiliates' emp (BSOBA), rel. domestic emp (TSR)
- SSIV

$$\Delta Z_i \equiv \sum_{c,s} \underbrace{\frac{Y_{i,c,s,t0}}{Y_{i,t0}}}_{\text{Share}} \cdot \underbrace{\Delta Z_{c,s}}_{\text{Shift}}$$

- $\Delta Z_{c,s}$: Country-sector specific FDI Regulation Index (OECD)
- $Y_{i,c,s,t}$: Payroll of firm i ' affiliate in country c , sector s , time t (BSOBA)
- Exposure-robust SEs

1st Stage: Looser Regulations Increase FDI

	Foreign Aff Emp Change/Baseline Domestic Emp			
	(1)	(2)	(3)	(4)
FDIRR exposure	0.78 (0.77)	-6.38*** (1.40)	-7.45*** (1.52)	-7.54*** (1.53)
Host GDP trend		-3.72*** (0.73)	-3.34*** (0.80)	-3.30*** (1.00)
Host RER trend			-1.14 (0.89)	-0.98 (1.21)
China exposure				-0.09 (0.69)
Obs.	2,669	2,669	2,669	2,669
First-stage F	1.05	20.93	24.17	24.27

Source: BSOBA, TSR, OECD

Reduced Form: Looser Regulations Increase Employment

	TSR parent domestic employment			
	(1)	(2)	(3)	(4)
FDIRR exposure	-0.11 (0.11)	-0.59*** (0.16)	-0.70*** (0.21)	-0.75*** (0.21)
Host GDP trend		✓	✓	✓
Host RER trend			✓	✓
China exposure				✓
Obs.	2,669	2,669	2,669	2,669

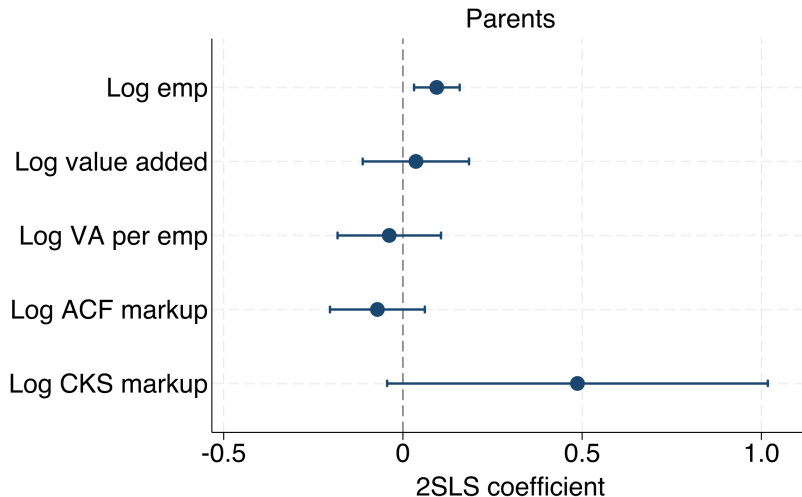
Source: BSOBA, TSR, OECD

2SLS Results: FDI Increases Parents' Employment

	TSR parent domestic employment			
	(1)	(2)	(3)	(4)
Affiliate employment change	-0.14 (0.20)	0.09*** (0.03)	0.09*** (0.03)	0.10*** (0.03)
Host GDP trend		✓	✓	✓
Host RER trend			✓	✓
China exposure				✓
Obs.	2,669	2,669	2,669	2,669
First-stage F	1.05	20.93	24.17	24.27

Source: BSOBA, TSR, OECD

2SLS Results: No Further Effect



Summary of Firm-level Empirical Results

- Revisit/replicate Kovak et al (2021) in Japan
- Confirm the key result (and the results are robust) across Firms

$$\text{corr}(\Delta\text{Offshoring}, \Delta\text{EMP}) > 0$$

Effects on Suppliers

Supplier-level Analysis: Specification

- Long-difference between 2010-2019
- Again, SSIV (long-difference); s supplier firms

$$\Delta \ln L_s = \beta \left[\sum_{i \in B(s)} \omega_{i,s} \cdot \Delta \text{OFS}_i \right] + \varepsilon_{s,t}$$

- SSIV: Suppliers' exposure to buyer-weighted FDI regulation

$$\Delta Z_s = \sum_{i \in B(s)} \omega_{i,s} \cdot \Delta Z_i, \quad \Delta Z_i \equiv \sum_{c,s} \frac{Y_{i,c,s,t0}}{Y_{i,t0}} \cdot \Delta Z_{c,s}$$

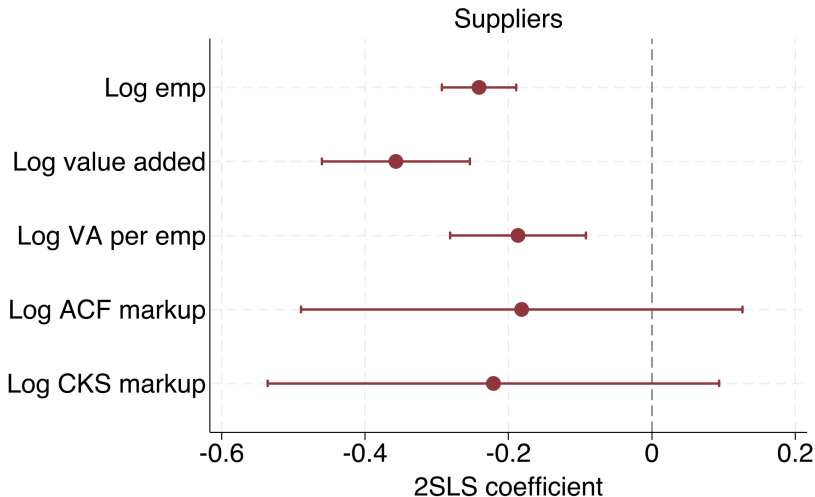
where $\omega_{i,s}$ is employment weight and ΔZ_i is our previous IV

2SLS: Buyers' FDI Decreases Suppliers' Employment

	Changes in supplier employment			
	(1)	(2)	(3)	(4)
Buyer FDI change	-0.18*** (0.03)	-0.24*** (0.03)	-0.24*** (0.03)	-0.24*** (0.03)
Host GDP IV		✓	✓	✓
Host RER IV			✓	✓
Own FDIRR control				✓
Obs.	491,176	491,176	491,176	491,176
First-stage F	222.88	111.77	85.43	83.75

Source: BSOBA, TSR, OECD

2SLS Results: Suppliers' Size and Productivity Decline



Simple Accounting: 2010-2019

- Before doing a full GE, try a simple PE accounting
 - "How many jobs increase/decrease per predicted offshoring, $\widehat{\Delta OFS}_i$?"

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- Direct effect (firm/parent-level regression coef $\widehat{\beta}^P$)

$$\widehat{\Delta L}^{\text{Direct}} = \sum_i L_{i,t0} \cdot \exp \left(\widehat{\beta}^P \cdot \widehat{\Delta OFS}_i \right) \approx 121K$$

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- Indirect effect (supplier-level regression coef $\widehat{\beta}^S$)

$$\widehat{\Delta L}^{\text{Indirect}} = \sum_s L_{s,t0} \cdot \exp \left(\widehat{\beta}^S \cdot \widehat{\Delta OFS}_s \right) \approx -520K$$

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$$\widehat{\Delta L^{\text{Indirect}}} = \sum_s L_{s,t0} \cdot \exp \left(\widehat{\beta^S} \cdot \widehat{\Delta OFS}_s \right) \approx -520K$$

- Net Effect: $\widehat{\Delta L^{\text{Direct}}} + \widehat{\Delta L^{\text{Indirect}}} \approx -399K$

Quantitative Model

Model Overview

Small open economy with two sectors (tradable vs non-tradable)

1. Tradable goods producers (=buyers) use domestic and foreign intermediates
 - Offshoring = substitution between domestic and foreign inputs
 - Heterogeneous in productivity, markup, and labor wedge
2. Non-tradable goods producers + Usual across-sector demand
3. Market clearing

Model 1/4. Tradable final-good producers

- Final assembly firm

$$Y^T = \left[M \int y(z)^{\frac{\varepsilon-1}{\varepsilon}} dG(z) \right]^{\frac{\varepsilon}{\varepsilon-1}}, \quad \varepsilon > 1,$$

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- Variety Production (Firm i with z_i); heterogeneous (z_i, μ_i, τ_i^L)

$$y_i = z_i \ell_i^\alpha x_i^{1-\alpha},$$

where x_i is an intermediate (next slide)

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- Heterogeneous Markup (μ_i)

$$p_i = \mu_i mc_i.$$

where marginal costs embed labor wedges τ_i^L too

$$mc_i = \frac{C_\alpha}{z_i} \left[w(1 + \tau_i^L) \right]^\alpha (P_i^x)^{1-\alpha}, \quad C_\alpha \equiv \alpha^{-\alpha} (1-\alpha)^{-(1-\alpha)}.$$

Model 2/4: Intermediate Input Choices: Offshoring

- Non-FDI firms source only domestically: $x_i = x_i^D$
- FDI firms pay fixed costs f^F and can source from abroad:

$$x_i = \left[\omega^{1/\sigma} \left(x_i^D \right)^{\frac{\sigma-1}{\sigma}} + (1-\omega)^{1/\sigma} \left(x_i^F \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad 0 < \omega < 1, \quad \sigma > 1,$$

- Foreign inputs available at the exogenous price

$$p^F = \tau \cdot \overline{p^F},$$

where τ is an offshoring wedge (our policy wedge)

Model 3/4. Non-Tradable and Household Demand

- Non-tradable Production

$$Y^N = z^N \ell^N,$$

- Households' demand for sectoral goods

$$U(C^T, C^N) = (C^T)^{1-\eta} (C^N)^\eta,$$

s.t.

$$I = w\bar{L} + \Pi + \text{Reb} + T, \quad C^T = (1 - \eta)I, \quad p^N C^N = \eta I.$$

Model 4/4. Market Clearing and Equilibrium

- Domestic intermediate market $X^D = M \int x^D(z) dG$,
- Nontradable market $Y^N = C^N$.
- Domestic labor market

$$\underbrace{M \int \ell(z) dG}_{\equiv L^P} + \underbrace{f^F M (z^*)^{-\theta}}_{\equiv L^{HQ}} + \underbrace{\ell^S}_{\equiv L^S} + \underbrace{\ell^N}_{\equiv L^N} = \bar{L},$$

- An equilibrium consists of a wage, prices, and allocations such that
 - Producers solve profit max. (labor demand, offshoring, FDI entry)
 - Households maximize utility
 - Markets clear

Effects of Offshoring

Equilibrium Characterization 1/2: Input Substitution

Foreign input price falls with τ

$$\tau \downarrow \implies p^F \downarrow$$

For offshoring firms, total intermediate price falls

$$d \log P^X = s_F d \log p^F < 0$$

Their domestic input share falls

$$d \log s_D = (\sigma - 1) s_F d \log p^F < 0$$

Therefore, when τ decreases,

$P^X \downarrow,$	$s_D \downarrow,$	$s_F = 1 - s_D \uparrow$
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Equilibrium Characterization 2/2: Firm-level responses

Let $\kappa \equiv (1 - \alpha)(\varepsilon - 1)$ and $\chi_i \equiv -\Delta \log p_i^F > 0$. For an incumbent offshoring firm,

$$\Delta \log L_i^P = \kappa s_F \chi_i$$

so that

$$p^F \downarrow \implies Y_i \uparrow, L_i^P \uparrow.$$

For domestic supplier input demand,

$$\Delta \log x_i^D = [\kappa - (\sigma - 1)] s_F \chi_i.$$

Hence domestic supplier demand falls iff

$$x_i^D \downarrow \iff \sigma - 1 > \kappa.$$

Calibration: Two IV Moments, Two Elasticities

Block	Objects	Values	Discipline
Baseline	(s_F^0, q, η)	(0.113, 0.241, 0.171)	JIP/TSR; baseline clearing
IV targets	$(\theta_P^{IV}, \theta_S^{IV})$	(0.094, -0.236)	Parent and supplier 2SLS
Elasticities	(κ, σ)	(0.830, 3.913)	Analytic IV inversion
Production	$(\alpha, \varepsilon, \theta)$	(0.919, 11.236, 5.863)	Cost shares/tail moments
Wedge loadings	(ϕ_μ, ϕ_τ)	(0.358, 0.172)	Markup/MRPL-size slopes
Counterfactual	$\hat{\tau}_1$	0.842	Match +1.26 M affiliates

Quantitative Results

Exercise.

- Choose τ_1 to reproduce FDIRR-predicted increase of 1.26 M affiliate jobs
 - $\tau : 1 \rightarrow 0.842$
- then solve the general equilibrium.

Results:

- Employment (reallocation)
- Productivity
- Welfare

GE Reallocates Labor to Nontradables

Employment outcome	PE Simple accounting	GE counterfactual
Parent firms	+121K	
Suppliers	−520K	
Exposed sector	−399K	
Nontradables	—	
Total domestic	—	

GE Reallocates Labor to Nontradables

Employment outcome	PE Simple accounting	GE counterfactual
Parent firms	+121K	+75K
Suppliers	−520K	−763K
Exposed sector	−399K	−688K
Nontradables	—	+688K
Total domestic	—	0, fixed $\bar{L}$

GE Reallocates Labor to Nontradables

Employment outcome	PE Simple accounting	GE counterfactual
Parent firms	+121K	+75K
Suppliers	-520K	-763K
Exposed sector	-399K	-688K
Nontradables	-	+688K
Total domestic	-	0, fixed $\bar{L}$

- The exposed-sector net loss remains negative in GE
- Nontradables absorb the displaced exposed-sector labor

Aggregate TFP: Technology Gain, Reallocation Drag

- Tradable Productivity: +0.21 pp

$$\underbrace{\Delta \log A_T}_{+0.21pp} = \underbrace{\Delta \log A_T^*}_{\text{technology/selection } (+0.20pp)} + \underbrace{\Delta \log AE}_{\text{allocative efficiency } (+0.01pp)}$$

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- Aggregate Productivity

$$A = \frac{A_M L_E + A_{NT} L_{NT}}{\bar{L}}, \quad \frac{A_{NT}}{A_M} = 0.635.$$

$$\Delta L_E = -688K, \quad \Delta L_{NT} = +688K.$$

$$\Delta \log A \approx \underbrace{+0.076}_{A_T \text{ gain}} \underbrace{-1.092}_{\text{reallocation to } A_{NT}} \underbrace{-0.005}_{\text{cross term}} \approx -1.021 \text{ pp}$$

Welfare Decomposition

$$\begin{aligned}
 d \log W \cdot I = & \underbrace{-X^F dp^F}_{\text{direct price fall}} \\
 & + \underbrace{M \int (\mu_i - 1) [w d\ell_i + p^D dx_i^D + p^F dx_i^F] dG_i}_{\text{markup reallocation}} \\
 & + \underbrace{M \int \mu_i \tau_i^L w d\ell_i dG_i}_{\text{MRPL/labor-wedge reallocation}} + \underbrace{S(z^*) dn_F}_{\text{FDI selection}}.
 \end{aligned}$$

Welfare (work-in-progress)

Component	pp
Direct import-price	+0.121
Markup reallocation	+0.124
Labor-wedge/MRPL reallocation	+2.842
FDI selection	+0.024
Total	+3.111

- Labor-wedge/MRPL reallocation: employment increases in high MRPL firms
- Currently missing: distortions in non-tradables, limited sectoral mobility...
 - so the welfare numbers are work-in-progress

Conclusion

- Offshoring **decreases** (exposed sector) employment at the **aggregate level**
 - opposite to the **cross-firm** sign

$$\text{corr}(\Delta\text{Offshoring}, \Delta\text{EMP}) > 0$$

- Caution for extrapolating micro evidence to macro questions
- Measured aggregate (domestic) TFP decreases
- Welfare improves (work-in-progress)

Appendix

2SLS: Related Suppliers Survive

	Changes in supplier employment			
	(1)	(2)	(3)	(4)
Parent-child buyer FDI	0.63 (0.53)	0.54*** (0.20)	0.27*** (0.10)	0.27*** (0.10)
Other buyer FDI	-0.20*** (0.03)	-0.26*** (0.03)	-0.27*** (0.03)	-0.27*** (0.03)
Split host GDP IVs		✓	✓	✓
Split host RER IVs			✓	✓
Own FDIRR control				✓
Obs.	491,176	491,176	491,176	491,176
First-stage F, parent-child	5.91	6.74	7.23	7.96
First-stage F, other buyers	127.73	65.01	48.02	47.31

Source: BSOBA, TSR, OECD